

My current research focuses on Agentic AI, with an emphasis on developing more effective and efficient ways to harness large language models as reliable, production-ready tools. In addition, I have research interests and experience in several other AI areas, including reinforcement learning, LLM post-training, and generative model, where I have also made research contributions.

EDUCATION

University at Buffalo, SUNY Department of Computer Science and Engineering	2024 – Present
Ph.D. Student in Computer Science GPA: 4.00 Advisor: Prof. Venu Govindaraju	
Syracuse University College of Engineering & Computer Science	2021 – 2023
M.S. in Computer Science GPA: 3.86	

PUBLICATIONS

- **Xiao Wang***, Lu Dong*, Sahana Rangasrinivasan, Ifeoma Nwogu, Srirangaraj Setlur, Venu Govindaraju. "AutoMisty: A Multi-Agent LLM Framework for Automated Code Generation in the Misty Social Robot." *International Conference on Intelligent Robots and Systems (IROS 2025 Oral)*.
- **Xiao Wang***, Lu Dong*, Ifeoma Nwogu, Srirangaraj Setlur, Venu Govindaraju. "MistyPilot: Enabling Social-Robot Control through Multi-Agent LLM Skill Orchestration." *Proceedings of the European Conference on Computer Vision Workshops (ECCVW 2026)*.
- **Xiao Wang***, Lu Dong*, Ifeoma Nwogu, Srirangaraj Setlur, Venu Govindaraju. "InterventionLens: A Multi-Agent Framework for Detecting ASD Intervention Strategies in Parent-Child Shared Reading." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2026)*.
- **Xiao Wang***, Lu Dong*, Mark G Frank, Srirangaraj Setlur, Venu Govindaraju, Ifeoma Nwogu. "ConfusionBench: An Expert-Validated Benchmark for Confusion Recognition and Localization in Educational Videos." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2026)*.
- **Xiao Wang***, Lu Dong*, Jingchen Sun, Ifeoma Nwogu, Srirangaraj Setlur, Venu Govindaraju. "MistyPilot: An Agentic Fast-Slow Thinking LLM Framework for Misty Social Robots." *IROS 2026, under review*.
- **Xiao Wang***, Lu Dong*, Srirangaraj Setlur, Venu Govindaraju, Ifeoma Nwogu. "Ig3D: Integrating 3D Face Representations in Facial Expression Inference." *Proceedings of the European Conference on Computer Vision Workshops (ECCVW 2024)*.
- Bhavin Jawade, Alexander Stone, Deen Dayal Mohan, **Xiao Wang**, Srirangaraj Setlur, Venu Govindaraju. "ProxyFusion: Face Feature Aggregation Through Sparse Experts." *The 2024 Conference and Workshop on Neural Information Processing Systems (NeurIPS 2024)*.
- Lu Dong, **Xiao Wang**, Ifeoma Nwogu. "Word-Conditioned 3D American Sign Language Motion Generation." *The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP 2024)*.
- Lu Dong, Lipisha Nitin Chaudhary, Fei Xu, **Xiao Wang**, Mason Lary, Ifeoma Nwogu. "SignAvatar: Sign Language 3D Motion Reconstruction and Generation." *The 18th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2024)*.

PROFESSIONAL EXPERIENCE

ByteDance / TikTok	May 2026 — Aug 2026
Research Scientist Intern, Applied Machine Learning – Enterprise (Evolve Agent)	San Jose, CA
• Developed algorithms for failure-signal analysis, variant selection, and Gene/Rendering error routing. • Proposed SkillAtom, a graph-based representation that decomposes natural-language Skills into traceable components for diagnosis and modification. • Developed Atomed Skill Evolve, which automatically optimizes Skills from strong-weak model execution traces; improved Haiku 4.5 task success from 14.1% to 45.3% across 64 tasks without changing model parameters.	
Microsoft (via Centific Global Solutions, Inc.)	May 2025 — Aug 2025
PhD Research Intern, MSAI Reasoning & Language Understanding (M365 Copilot Researcher)	USA
• Developed a four-stage multi-agent meeting simulation pipeline spanning scene planning, character modeling, directed interaction, and post-production. • Built automated fact-evaluation and feedback mechanisms to detect factual deviations and improve generation quality. • Generated hundreds of high-fidelity synthetic meeting transcripts, typically 40–60 minutes and 150+ turns, for downstream internal training and research.	